

Session 19

Dart Inheritance & Polymorphism Assignment

Objective: Learn inheritance, method overriding, static members, the super and this keywords, and polymorphism in Dart.

Task 1: ProductUser Class

Description: Create a ProductUser class with name, email, and a displayInfo() method.

```
class ProductUser {
  static int totalUsers = 0;
  String name;
  String email;

  ProductUser(this.name, this.email){
    totalUsers++;
  }

  void displayInfo() {
    print("Name: $name");
    print("Email: $email");
  }
}

void main() {
  ProductUser user =
    ProductUser("Jainish", "jainish@gmail.com");
  user.displayInfo();
}
```

Expected Output:

Name: Jainish

Email: jainish@gmail.com

Task 2: SellerUser Class

Description: Extend ProductUser and override displayInfo() using super.

```
class SellerUser extends ProductUser {
    String shopName;

    SellerUser(String name,String email,this.shopName)
        : super(name,email);

    @override
    void displayInfo() {
        super.displayInfo();
        print("Shop: $shopName");
    }
}
```

Expected Output:

Name: Jainish
Email: jainish@gmail.com
Shop: Tech Store

Task 3: Static Variable

Description: Track the total number of users using a static variable.

```
void main() {
    ProductUser("A","a@mail.com");
    SellerUser("B","b@mail.com","Shop");
    CustomerUser("C","c@mail.com");

    print(ProductUser.totalUsers);
}
```

Expected Output:

3

Task 4: CustomerUser Class

Description: Create a CustomerUser class with placeOrder(productName).

```
class CustomerUser extends ProductUser {
    CustomerUser(String name,String email)
        : super(name,email);

    void placeOrder(String productName) {
        print(
            "Order placed for $productName by ${this.name}"
        );
    }
}
```

Expected Output:

Order placed for Laptop by Jainish

Task 5: Polymorphism

Description: Write processUser(ProductUser user) to demonstrate polymorphism.

```
void processUser(ProductUser user) {
    user.displayInfo();
}

void main() {
    SellerUser seller =
        SellerUser("Raj", "raj@mail.com", "Mobile Hub");
    CustomerUser customer =
        CustomerUser("Amit", "amit@mail.com");

    processUser(seller);
    processUser(customer);
}
```

Expected Output:

Seller details printed, then customer details printed through the same function.

Conclusion: This session demonstrates inheritance, method overriding, static variables, the super and this keywords, and runtime polymorphism using a common base class in Dart.